TMDB Movie Analysis

By: Molo Munyansanga

Initial Summary

- **Profit**: Every movie made over 1 billion USD in profit. **Avatar** made the most profit (2.69B USD) and had the highest ROI, earning 12.3 times its budget.
- **Revenue**: **Avengers: Endgame** had the highest revenue (2.94B USD), just ahead of **Avatar** (2.92B USD).
- **Budget**: The highest budgets went to action/sci-fi franchise movies, led by **Avengers: Endgame** (356M USD). This is probably due to the extensive use of special effects and CGI.
- **Popularity**: The Avengers movies have been among the most popular and highest rated movies. The Avengers Collection is also the most successful franchise (7.9B USD from 4 movies).
- **Directors**: **James Cameron** earned the most total revenue (5.19B USD from Avatar and Titanic).

Note: The dataset only contained 18 rows, all of them blockbuster movies, so this may not be representative of the whole movie industry.

Methodology

1. **Data Collection**: The project used movie data from the TMDb (The Movie Database) API, focusing on a selected dataset of 18 high-grossing movies. Requests use a timeout and retry temporary failures.

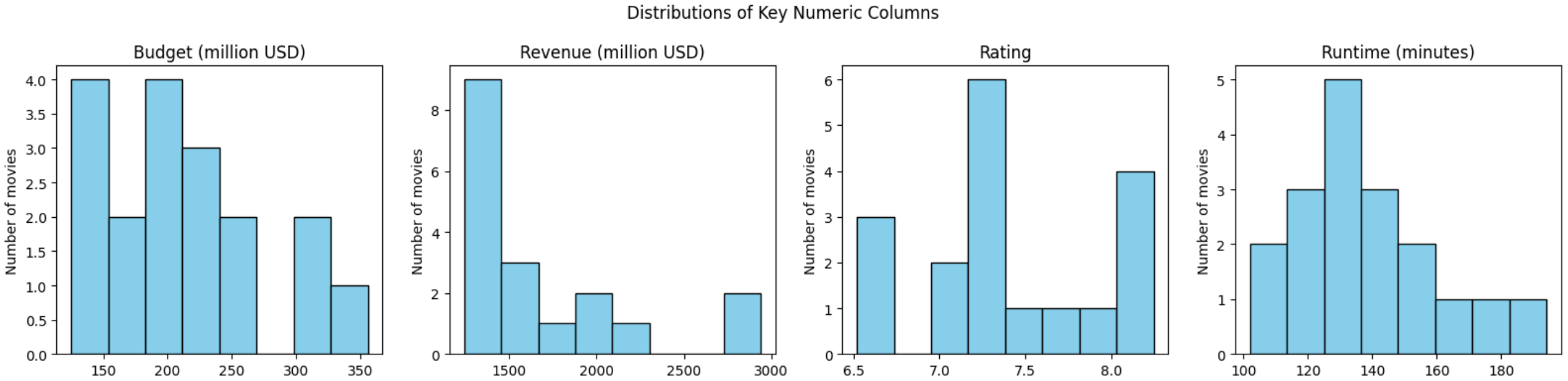
2. **Data Cleaning and Preparation**: Python and Pandas were used to flatten the nested JSON columns, fix data types, replace unrealistic values with NaN, convert budget and revenue to million USD, and remove invalid rows.

3. **Exploratory Data Analysis**: Summary statistics, distributions, genre counts and correlations were used to understand the dataset before the main analysis.

4. **Data Analysis**: Movies were ranked by revenue, budget, profit, ROI, votes, rating and popularity, and franchises, standalone movies and directors were compared.

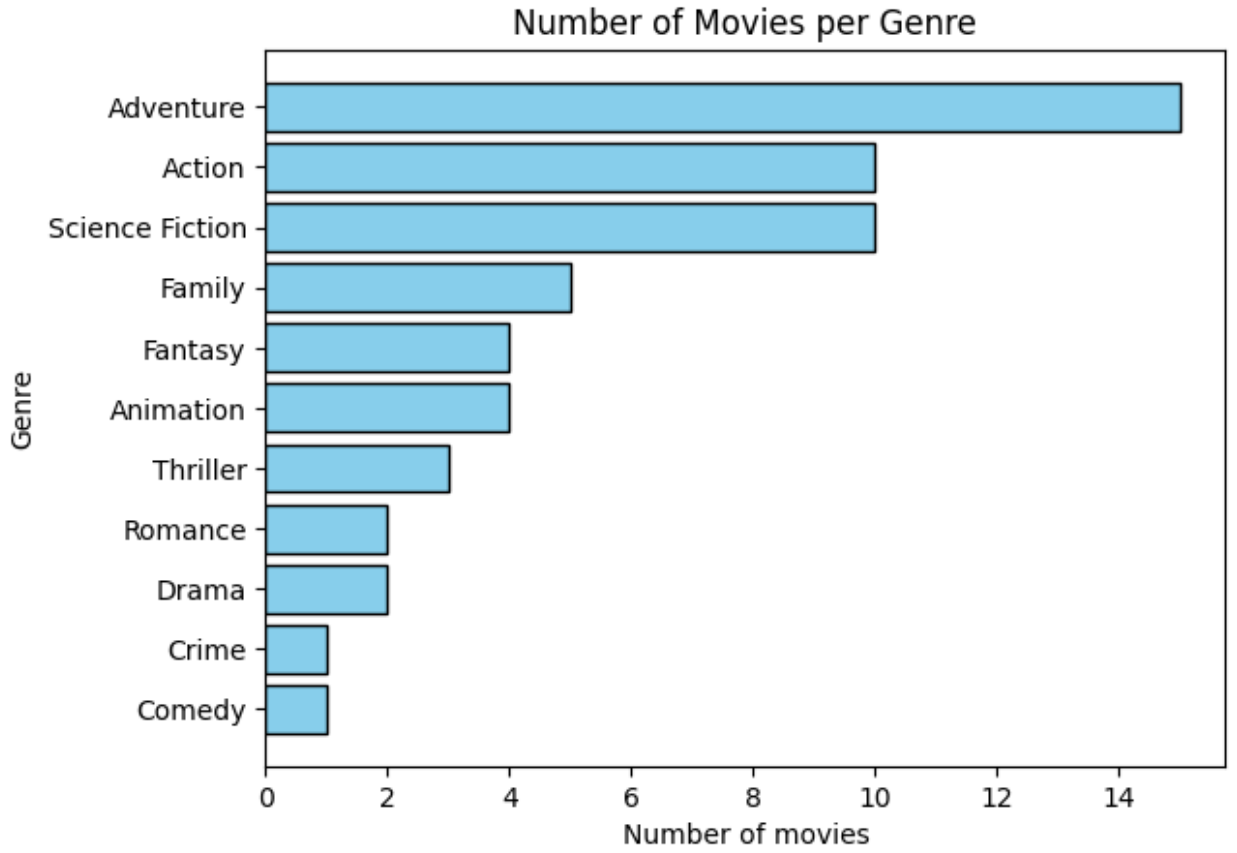
5. **Data Visualization**: Matplotlib and Seaborn were used to create histograms, a heatmap, scatter plots with trend lines, horizontal bar charts, and line charts.

Distributions of Key Columns



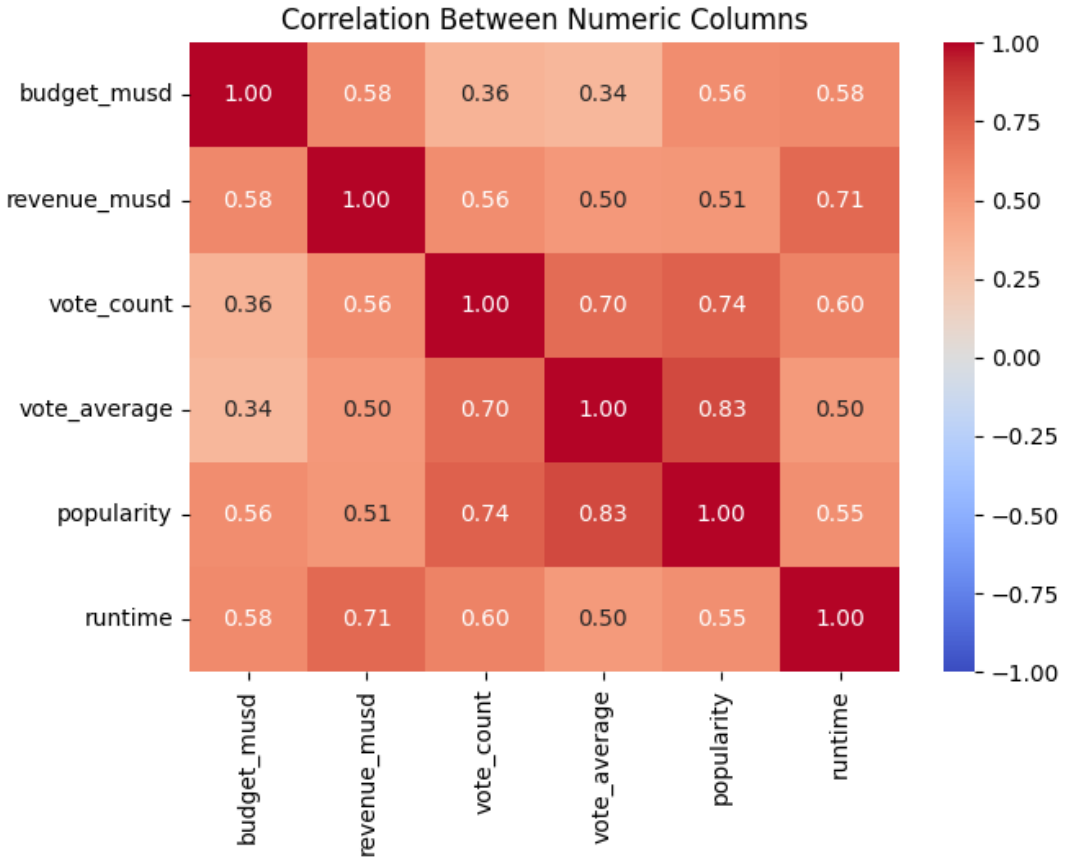
Half the movies earned between 1.2B and 1.45B USD, with two outliers near 2.9B USD. Most are rated 7.2 to 7.4 and run 120 to 140 minutes.

Movies per Genre



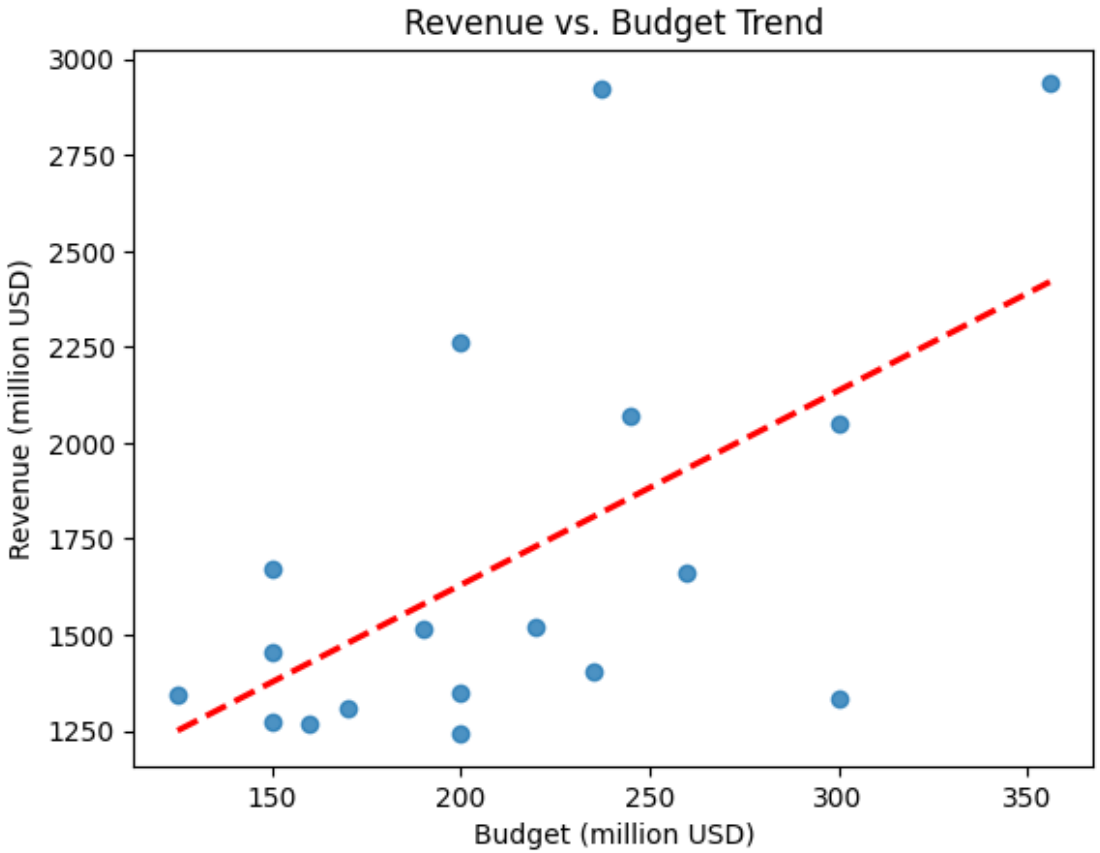
Adventure is the most common genre (15 of 18 movies), followed by Action and Science Fiction (10 each).

Correlations



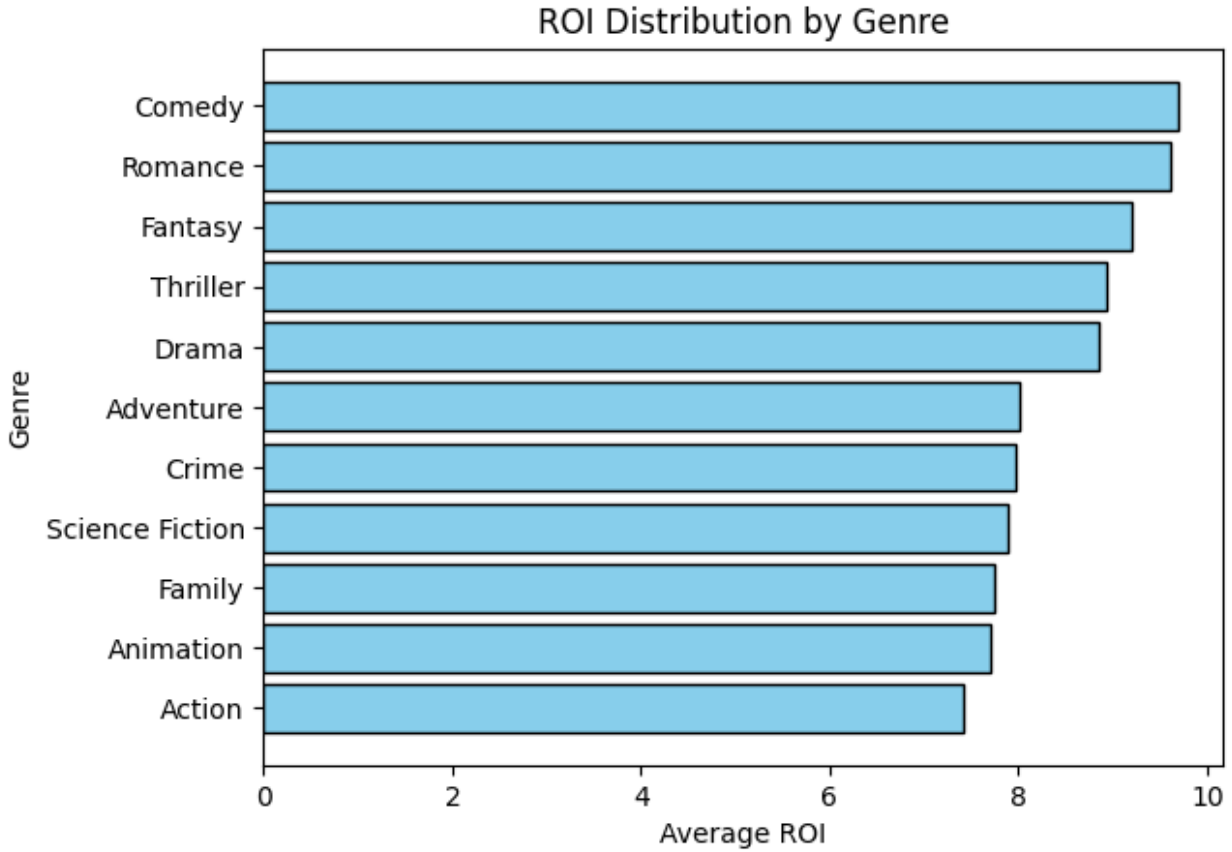
Popularity and rating are strongly linked (0.83), while budget is only moderately linked to revenue (0.58).

Revenue vs. Budget



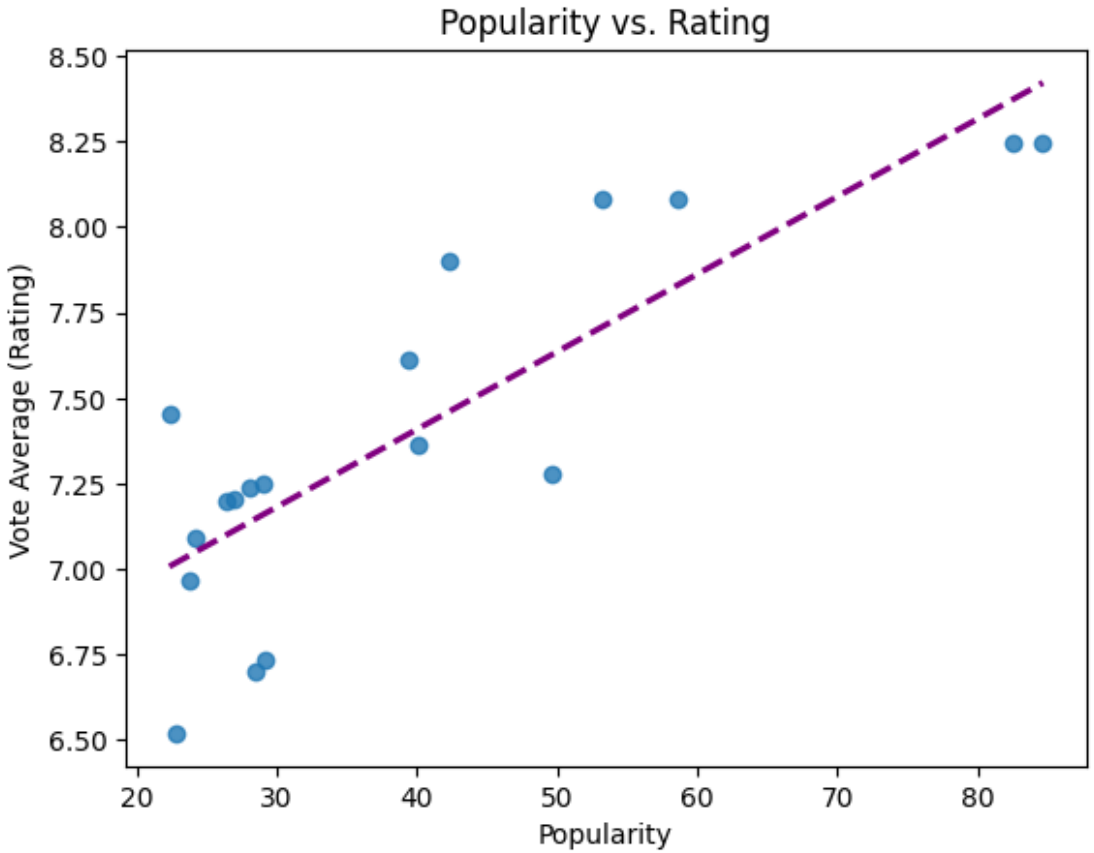
Revenue tends to rise with budget, but the points are widely scattered: Avatar earned about 2.9B USD on a 237M USD budget.

ROI Distribution by Genre



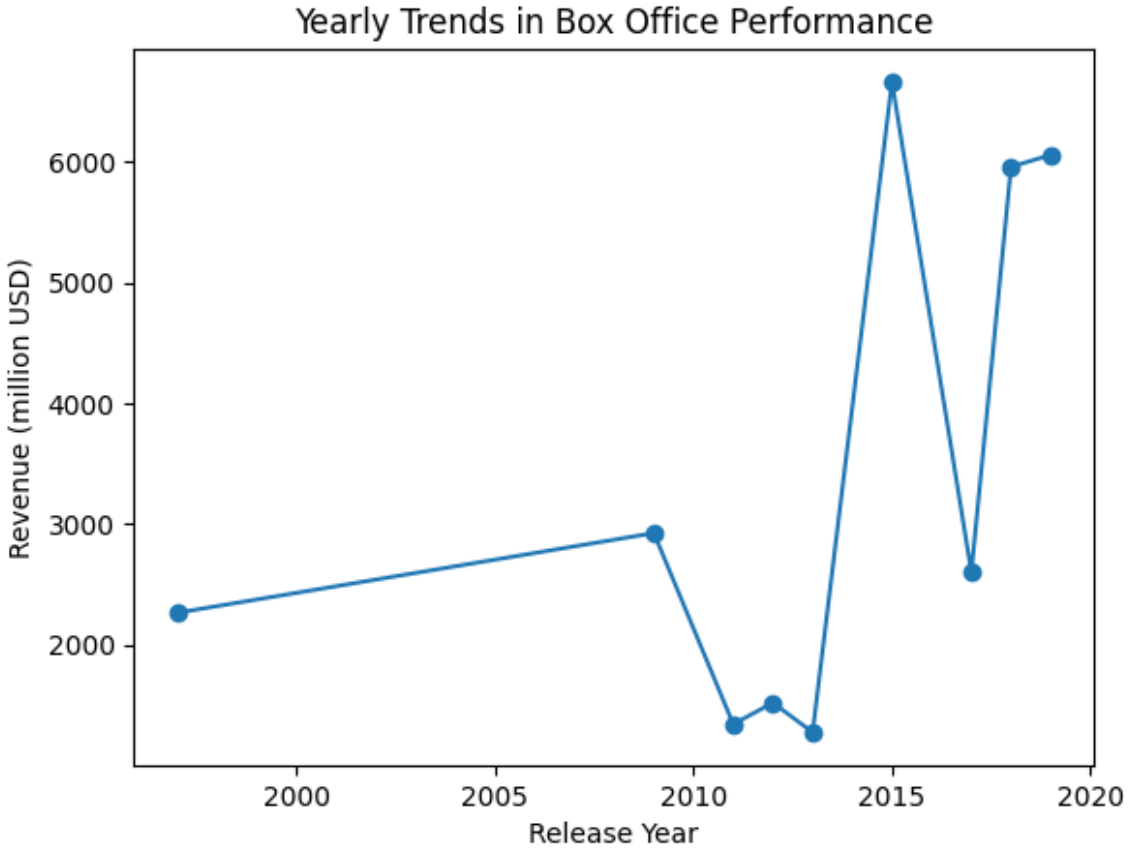
Comedy and Romance have the highest average ROI and Action the lowest, but Comedy and Romance include only 1 to 2 movies each.

Popularity vs. Rating



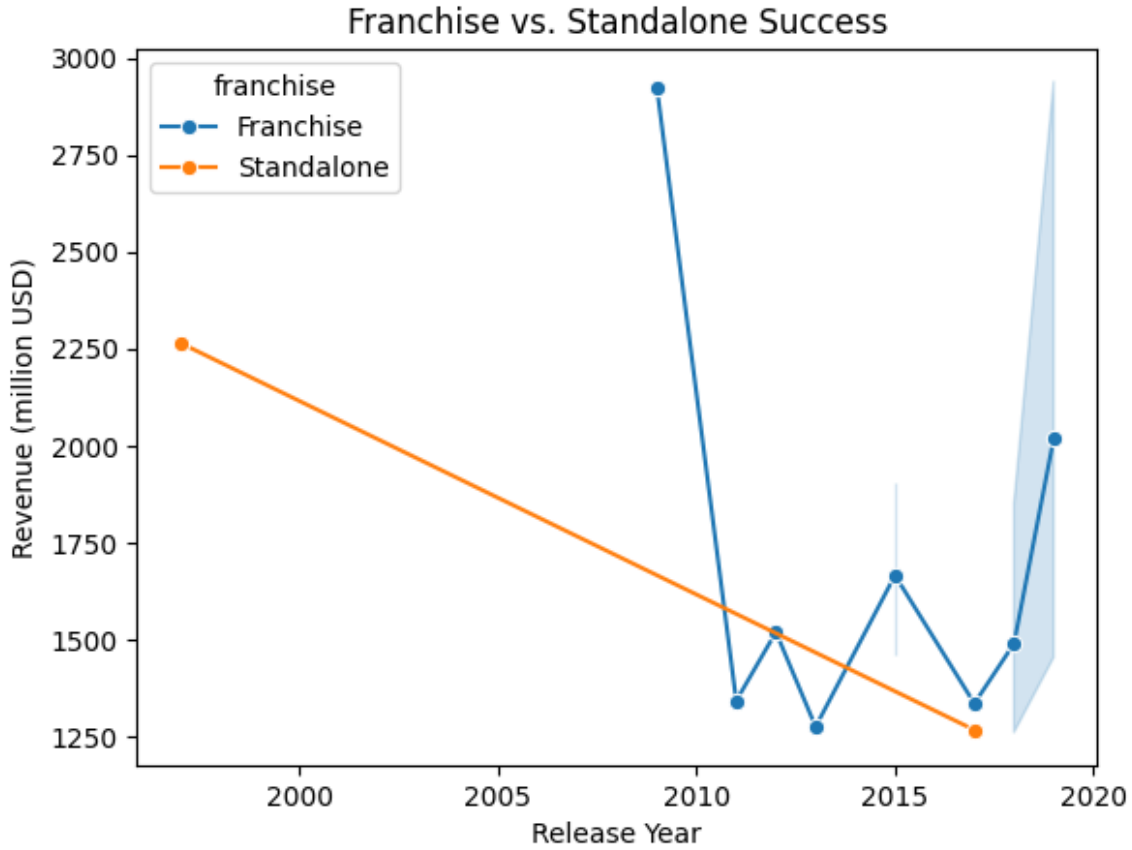
More popular movies tend to have higher ratings. The two most popular movies (the latest Avengers films) are also rated highest.

Yearly Trends in Box Office Performance



Revenue peaks in 2015 and 2018 to 2019, mainly because more of the dataset's movies were released in those years.

Franchise vs. Standalone Success



The standalone line connects only two movies (Titanic and Beauty and the Beast), so it does not show a real trend.

Conclusions

1. More data needs to be collected for a more concrete movie analysis. All 18 movies are blockbusters, so the results describe top-grossing movies rather than movies in general.

2. **Titanic** had the third-highest profit (2.06B USD), behind only **Avatar** and **Avengers: Endgame** and ahead of movies like **Star Wars** and **Harry Potter**. This is more impressive considering it is a standalone movie and not a fantasy/sci-fi movie. A deeper analysis may be needed in order to figure out how to reproduce this success.

3. Franchise movies had bigger budgets (218M vs. 180M USD on average) and were more popular, but the standalone movies had slightly higher mean revenue and ROI. With only 2 standalone movies, it is not yet clear whether franchises do better. This may be an interesting avenue to explore with more data.

4. A bigger budget does not guarantee more revenue: budget is only moderately linked to revenue, while popularity and rating are closely linked.